

Mohamed Elgazar

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Summary

A highly motivated AI engineer with hands-on experience in machine learning, deep learning, and data-driven solutions. Skilled in developing end-to-end AI applications, from model training to deployment using modern tools and frameworks. Having experience in full-stack web development to build intuitive user interfaces and seamless integrations. Passionate about solving real-world problems through innovative AI approaches, with a strong foundation in Python, data analysis, and cloud deployment. Proven ability to work collaboratively in teams and deliver impactful, production-ready projects.

Education

Tanta University, BS in AI Engineering Oct 2022 – May 2027

- GPA: 3.96/4.0
- **Coursework:** Machine Learning, Deep Learning, Neural Networks, Data Structures, Operating Systems, Computer Architecture, Databases, Microprocessors & Microcontrollers.

Experience

Full-Time Developer at Meshkode – Remote Aug 2025 - Feb 2026

- Gained strong foundations in frontend development using HTML, CSS, JavaScript, and Angular, focusing on scalable and maintainable user interfaces.
- Applied modern frontend testing practices with Jest, React Testing Library (RTL), and Playwright to ensure high code quality.
- Developed hands-on experience in AI systems, including LangChain, LangGraph, prompt engineering, vector databases, and RAG.
- Worked with agent-based AI workflows and developer tools such as Agents.md and Continue Dev .
- Built and maintained development Environments using GitHub Actions, Temporal workflows, and monorepo architectures.
- Gained practical experience with graph databases using Neo4j and Cypher for modeling and querying relations.

Machine Learning Engineer (Depi Training Program), Microsoft – Remote Jun 2025 – Jan 2026
Under the supervision of EYouth Co. *AI & Data Science Track*

- Completed a hands-on training program focused on Microsoft's AI and Data Science tools and pipelines.
- Built and deployed ML models using supervised learning algorithms to solve classification problems.
- Gained practical experience with Azure ML, data preprocessing, and model evaluation techniques.

Projects

MeshKode – AI-Powered Codebase Intelligence Agent [Github Repo](#)

- Built an agentic system that ingests any GitHub repository, constructs a semantic graph of the entire codebase, and enables an AI agent to understand, navigate, and modify code — inspired by Google's CodeWiki approach.
- Designed a clustering pipeline that groups code symbols and files into semantic units, giving the agent a structured mental model of large monorepos without exceeding context limits.
- Implemented a multi-step agentic workflow where the agent uses codebase knowledge to autonomously plan and apply refactoring, feature additions, and bug fixes across the project.
- Integrated graph-based code analysis using Neo4j and pgvector for semantic search, with Qdrant Cloud for embedding storage and retrieval.
- **Tools Used:** Python, NestJS, Angular, Shell, Nx, Neo4j, pgvector, Qdrant Cloud, Temporal, Sipher, Scip, ts-morph, Joren, Continue.dev, Monorepos.

EduBot – Arabic AI Tutoring Assistant

[Live Demo](#) | [Github Repo](#)

- Developed an Arabic-first AI tutoring system that converts educational PDFs into an interactive learning assistant using Retrieval-Augmented Generation (RAG).
- Implemented both standard and agentic RAG pipelines with LangGraph, enabling multi-step reasoning, memory usage, and web search.
- Built a curriculum-aware vector retrieval system using Qdrant and multilingual embeddings, filtered by subject, grade, and term.
- Designed a student memory module to personalize explanations and track learning progress across conversations.
- Integrated Arabic voice capabilities with speech-to-text and text-to-speech for hands-free interaction.
- **Tools Used:** Python, Flask, LangChain, LangGraph, Qdrant, Hugging Face, Groq API, React, Express, Docker.

Balagh AI – Islamic Outreach Platform

[Live Demo](#)

- Co-developed a multi-role platform connecting Muslim preachers, non-Muslims, Islamic organizations, and platform admins, currently deployed in Kuwait.
- Built a RAG-powered chatbot for preachers to retrieve accurate Islamic answers from a curated knowledge base during conversations.
- Designed a 24/7 chatbot for non-Muslim users to ask questions freely and get instant responses before engaging with a preacher.
- Developed an analytics chatbot for admins and supervisors connected directly to the database to query platform statistics and activity reports.
- **Tools Used:** React, Python, FastAPI, Qdrant, DeepSeek.

Brain Tumor Classification and Care

[Github Repo](#)

- Improved a CNN-based model to classify brain tumors from MRI scans with high accuracy.
- Compared performance of TensorFlow and PyTorch implementations on the same dataset.
- Integrated a care recommendation system and built an LLaMA 3-powered chatbot for post-diagnosis assistance.
- Designed a user-friendly interface to visualize predictions, confidence scores, and interact with the care assistant.
- **Tools Used:** Python, TensorFlow, PyTorch, LLaMA 3, HTML/CSS, Flask, Open WebUI.

Expenses Tracker Web App

[Github Repo](#)

- Developed a full-stack web application to manage income and expenses with JWT-based authentication.
- Implemented features for adding, editing, deleting, and categorizing transactions, with real-time visualization.
- Built a responsive and accessible UI ensuring smooth user experience across devices.
- Designed system architecture diagrams using Excalidraw and created UI/UX mockups in Figma.
- **Tools Used:** React, Next.js, Tailwind CSS, Node.js, Express.js, MongoDB, Chart.js, JWT, Figma, Excalidraw.

Skills

Programming Languages: Python, C++, C#, JavaScript, TypeScript, SQL, HTML/CSS, Tailwind CSS

AI & Machine Learning: LangChain, LangGraph, LangSmith, Llamaindex, Scikit-learn, TensorFlow, PyTorch, OpenCV, NumPy, Pandas, CNNs, transfer learning, model evaluation, clustering, regression, NLP

Development & Deployment: React, Node.js, MongoDB, Flask, FastAPI, REST APIs, Docker, CI/CD, socket programming, client-server architecture

Soft Skills: Problem Solving, Critical Thinking, Adaptability, Creativity, Initiative, Time Management, Client Communication, Teamwork

Languages

Arabic: Native

English: Professional Proficiency